



Virulence gene profile and antimicrobial resistance of non-typeable *Streptococcus suis* isolated from clinically healthy and diseased pigs from North East India

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Abstract

The present study was conducted to investigate the virulence gene profile and antimicrobial resistance of non-typeable *Streptococcus suis* isolates circulating in pigs of North East India. Fifty-two non-typeable *S. suis* isolates from clinically healthy and diseased pigs were screened by using PCR for the presence of the muramidase-released protein (*mrp*), extracellular factor (*epf*), hemolysin suilysin (*sly*), arginine deiminase (*arcA*), and glutamate dehydrogenase (*gdh*) genes. Five different virulence gene profiles were observed and the most predominant virulence gene profile found in healthy pigs was *mrp*[−] + *sly*[−] + *arcA*[−] + *gdh*⁺ + *epf*[−] whereas the most predominant virulence gene profile recorded in diseased pigs was *mrp*⁺ + *sly*[−] + *arcA*⁺ + *gdh*⁺ + *epf*[−]. Significantly lower carrier rate of *mrp*⁺ + *sly*[−] + *arcA*⁺ + *gdh*⁺ + *epf*[−] virulence gene profile was observed among the isolates from healthy pigs compared to those from diseased pigs ($P < 0.05$). Antimicrobial resistance patterns of the *S. suis* isolates revealed fourteen resistance groups (R1 to R14) where 88.46% isolates showed multi-drug resistance. The most predominant resistance pattern observed was CD-COT-E-TE. This is perhaps the first study reporting virulence gene profile and antimicrobial resistance of non-typeable *S. suis* isolates from pigs in North East India. The occurrence of relatively high levels of resistance of *S. suis* to some antimicrobials (e.g. macrolides, tetracyclines, and sulphonamides) as observed in the present study may represent a human health concern.

Introduction

Streptococcus suis has emerged as an important swine pathogen in most countries with an intensive swine production (Berthelot-Herault et al. 2005). This organism is associated with a variety of diseases including meningitis, polyarthritis, polyserositis, endocarditis, septicaemia, pneumonia, and sudden death, leading to significant economic losses in pig farms (Wei et al. 2009). It is also an emerging zoonotic pathogen which poses a potential risk for those working closely with pigs such as pig farmers, slaughter house workers, persons transporting pork, meat inspectors, and butchers and has received increasing attention worldwide (Gottschalk et al. 2010). Human infection results from

prolonged or repeated close contact with infected pigs or their products. Epidemiological surveillance has identified a number of risk factors for human infection including the consumption of undercooked pork (Nghia et al. 2011). During the past few years, the number of human *S. suis* infections reported worldwide has increased significantly with most cases reported in Asia. It has been shown that *S. suis* is the first cause of adult meningitis in Vietnam, the second in Thailand and the third most common cause of community-acquired bacterial meningitis in Hong Kong (Suankratay et al. 2004; Hui et al. 2005; Ip et al. 2007; Mai et al. 2008).

Serotyping is one of the important tools for diagnosis and is used in epidemiological studies of *S. suis*. As per Kerdin et al. (2014a), with removal of 4 serotypes (20, 22, 26, and 33) from original 33 serotypes, a total of 29 was considered to be true *S. suis* serotypes. Serotype 2 is the most prevalent serotype in human as well as diseased pigs. Many serotypes from pigs and serotypes 1, 4, 5, 14, 16, and 24 have also been reported from human cases (Gottschalk 2011; Liu et al. 2013). These serotypes are classified on the basis of the antigenic differences in their capsular polysaccharides (CPSs) and the coagglutination test. The *S. suis cps* gene clusters

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of these serotypes have been sequenced, and the *cps* gene clusters were shown to be diverse among different serotypes (Okura et al. 2013). However, non-typeable *S. suis* strains from both healthy and diseased pigs have been frequently reported in many studies (Wang et al. 2012; Gottschalk et al. 2013; Okura et al. 2014; Zheng et al. 2017). *S. suis* infection in humans caused by non-typeable strain has also been reported (Kerdsin et al. 2014b). As some *S. suis* strains do not express the CPS and hence cannot be serotyped using antisera, serotyping based on molecular techniques has been proposed (Zheng et al. 2017). Since the *S. suis* CPS is synthesized by the Wzx/Wzy pathway in the CPS locus, *wzy* genes have been demonstrated to be serotype-specific (Okura et al. 2013). Thus, high-throughput capsular gene typing systems based on serotype-specific *wzy* genes have become attractive alternatives/complement to the existing serological tests (Okura et al. 2014). However, even with the use of multiplex PCR tests, non-serotypeable strains are still commonly isolated from both clinically healthy and diseased animals (Okura et al. 2014; Qiu et al. 2016).

Factors such as SLY (encoded by the *sly* gene), MRP (*mrp* gene), and EF (*epf* gene) are virulence markers that have been used in elaborated genotypic and phenotypic schemes to predict the virulence of a given *S. suis* strain (Gruening et al. 2006; Baums and Valentin-Weigand 2009). It is now included in routine diagnosis of *S. suis* in several laboratories (Baums and Valentin-Weigand 2009).

Antimicrobial drugs have long been applied as therapeutic and/or prophylactic medication for *S. suis* infection in pigs and humans. However, the imprudent use of antimicrobial drugs results in serious antimicrobial resistance (AMR) that has become a global problem in recent years (Yongkiettrakul et al. 2019). Increasing cases of AMR in *S. suis* isolated from pigs and humans were observed from various countries in Asia, Europe, and North America (Zhang et al. 2008; Varela et al. 2013). Such cases increase the risks of therapeutic failure in animals and humans. Among antimicrobials examined, a high rate of resistance to lincosamides, macrolides, sulphonamides and tetracyclines (TETs) has been frequently reported (Varela et al. 2013). The aim of the present study is to investigate the virulence gene profile and antimicrobial resistance of non-typeable *S. suis* isolated from clinically healthy and diseased pigs from North East India.

Materials and methods

Sample collection

A total of 52 *S. suis* isolates recovered from the pigs of North East India during 2014–2018 and found negative for all the 33 serotypes in PCR (as described by Liu et al. 2013)

were included in this study. Of these 52 isolates, 19 isolates were from nasal swabs of clinically healthy pigs and rest 33 were from diseased pigs [pneumonia (N = 16), arthritis (N = 11) and septicemia (N = 6)].

Confirmation of the isolates

The isolates were grown on Columbia agar plate (Himedia, Mumbai, India) supplemented with 5% bovine blood. The plates were incubated aerobically at 37 °C and inspected for growth after 24 h. Colonies showing green or brown decolorization around the colony were suspected as potential *S. suis*. The primary identification of the organism was made as per the method of Quinn et al. (1994). Further, the organism was confirmed as *S. suis* by PCR using primer specific to 16S rDNA (Marois et al. 2004). DNA templates were prepared from 2 to 3 colonies of *S. suis* grown on blood agar and washed by suspending in 1 mL of distilled water (DW), then centrifuged at 12,000 rpm for 5 min. The supernatant was discarded and 100 µL of DW was added, then boiled for 10 min and quickly cooled in an ice box. Finally, centrifuged at 12,000 rpm for 5 min, the supernatant was then collected and used as template for PCR.

Detection of virulence-associated genes

DNA template from each isolate was prepared as described above. Amplification of virulence-associated genes, suilysin (*sly*), muramidase released protein (*mrp*), arginine deiminase (*arcA*), and extracellular protein factor (*epf*), was achieved by monoplex PCR assays using the sequence of primers and conditions described previously by Silva et al. (2006). The gene encoding for the enzyme glutamate dehydrogenase (*gdh*) was detected as per the method of Okwumabua et al. (2003). Positive and negative control was also used in the study. The positive control used was well characterized *S. suis* isolate (positive for *mrp*, *epf*, and *sly* genes) maintained at the Animal Health Laboratory, National Research Centre on Pig, Indian Council of Agricultural Research, Rani, Guwahati, Assam, India whereas the negative control was nuclease free water. Details of all oligonucleotide primers used for the detection are listed in Table 1.

Antimicrobial susceptibility testing

Antimicrobial susceptibility testing was performed by the disk diffusion method using 12 commercially available antimicrobials disks as per the method of the Clinical and Laboratory Standards Institute guidelines (CLSI 2008). The antimicrobial agents used in this study were: penicillin G (P), 10 IU; ampicillin (AMP), 10 µg; tetracycline (TE), 30 µg; streptomycin (HLS), 300 µg; gentamicin (HLG), 120

Table 1 Oligonucleotide primers sequences and expected PCR products

Gene	Nucleotide (5'-3')	Expected PCR product (bp)
16S rDNA	CAGTATTTACCGCATGGTAGATAT GTAAGATACCGTCAAGTGAGAA	294
<i>epf</i>	CGCAGACAACGAAAAGATTGA AAGAATGTCTTTGGCGATGG	744
<i>sly</i>	GCTTGACTTACGAGCCACAA CCGCGCAATACTGATAAGC	248
<i>mrp</i>	ATTGCTCCACAAGAGGATGG TGAGCTTTACCTGAAGCGGT	188
<i>arcA</i>	TGATATGGTTGCTGCTGGTC GGACTCGAGGATAGCATTGG	118
<i>gdh</i>	GCAGCGTATTCTGTCAAACG CCATGGACAGATAAAGATGG	688

µg; amikacin (AK), 30 µg; ciprofloxacin (CIP), 5 µg; enrofloxacin (EX), 10 µg; erythromycin (E), 15 µg; clindamycin (CD), 2 µg; chloramphenicol (C), 30 µg and co-trimoxazole (COT), 25 µg.

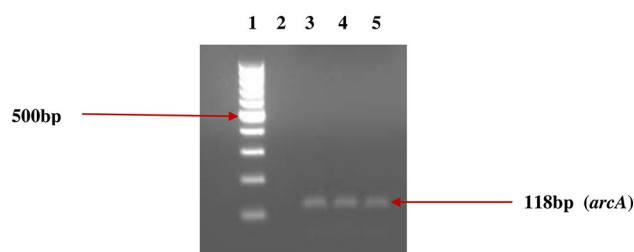
Statistical analysis

Analyses were performed with SPSS 13.0 software for Windows with a probability (P) value < 0.05 as statistically significant.

Results

Virulence-associated genes analysis

Fifty-two (52) non-typeable *S. suis* isolates obtained from the repository were screened for the presence of the *mrp*, *epf*, *sly*, *arcA*, and *gdh* genes. The distribution of virulence-associated genes is shown in Table 2. Arginine deiminase (Fig. 1), *sly* (Fig. 2) and *mrp* (Fig. 3) genes were present in

**Fig. 1** Detection of *arcA* gene of *S. suis* from pig by PCR. Lane 1: Molecular marker (100 bp DNA ladder), Lane 2: Negative control, Lane 3: Positive control, Lane 4–5: Test organism

5, 6, and 7 isolates, respectively from healthy pigs. All the isolates from both healthy and diseased pigs were positive for *gdh* gene (Fig. 4). It was also observed that *sly*, *mrp*, and *arcA* genes were present in 21, 23, and 27 isolates, respectively, from diseased pigs. All the isolates from both healthy and diseased pigs were negative for *epf* gene. Five different virulence gene profiles was observed among the isolates from healthy pigs and the most predominant virulence gene profile was *mrp*[−] + *sly*[−] + *arcA*[−] + *gdh*⁺ + *epf*[−]. Five different virulence gene profiles were also found among the isolates from diseased pigs and the most predominant virulence gene profile recorded was *mrp*⁺ + *sly*[−] + *arcA*⁺ + *gdh*⁺ + *epf*[−]. Significantly lower carrier rate of *mrp*⁺ + *sly*[−] + *arcA*⁺ + *gdh*⁺ + *epf*[−] virulence gene profile was observed among the isolates from healthy pigs compared to those from diseased pigs (*P* < 0.05).

Antimicrobial susceptibility testing

Fifty-two (52) non-typeable *S. suis* isolated under this study were tested for antimicrobial susceptibility test using 12 commonly used antimicrobials (Table 3). High frequency of resistance was observed for antimicrobials such as tetracycline (88.46%), clindamycin (84.61%), co-trimoxazole (82.69%), and erythromycin (78.84%). The most effective antimicrobial against *S. suis* was ampicillin followed by penicillin G whereas none of the

Table 2 Virulence gene profile of 52 non-typeable *S. suis* isolates from pigs

Virulence gene profile*	Number of <i>S. suis</i> isolates				Total
	Healthy pigs	Diseased pigs			
		Pneumonia	Arthritis	Septicaemia	
<i>mrp</i> ⁺ + <i>sly</i> ⁺ + <i>arcA</i> ⁺	2	2	2	1	7
<i>mrp</i> ⁺ + <i>sly</i> ⁺ + <i>arcA</i> [−]	3	2	3	1	9
<i>mrp</i> ⁺ + <i>sly</i> [−] + <i>arcA</i> ⁺	2	12	0	0	14
<i>mrp</i> [−] + <i>sly</i> ⁺ + <i>arcA</i> ⁺	1	0	6	4	11
<i>mrp</i> [−] + <i>sly</i> [−] + <i>arcA</i> [−]	11	0	0	0	11
Total	19	16	11	6	52

* All the isolates were positive of *gdh* gene and negative for *epf* gene



Fig. 2 Detection of *sly* gene of *S. suis* from pig by PCR. Lane 1: Molecular marker (100 bp DNA ladder), Lane 2: Negative control, Lane 3: Positive control, Lane 4–5: Test organism

antimicrobials was 100% effective against these isolates. Antimicrobial resistance patterns of the *S. suis* isolates are shown in Table 4. Fourteen resistance groups (R1 to R14) were observed in the present study where 88.46% isolates were multi-drug resistant. It has also been observed that 45 (86.53%) isolates were resistant to at least four antimicrobials. The most predominant resistance pattern observed was CD-COT-E-TE which accounted for 34.61% of the isolates.

Discussion

It was observed that out of 33 non-typeable *S. suis* isolates recovered from diseased pigs under this study, majority (48.48%) of the isolates were from cases of pneumonia. The potential causative role of this organism in respiratory diseases in pigs has been suggested by Messier et al. (2008). Reports of isolation of this bacterium from pigs suffering from arthritis and septicaemia are also frequent (Baums and Valentin-Weigand 2009; Wei et al. 2009). As in the present study, the isolation of non-typeable *S. suis* from pigs has been reported by several workers. Marois et al. (2007) reported that among the 406 *S. suis* isolates recovered from the 8 herds in France, non-typeable strains accounted for 237 (58%). Similarly, Okura et al. (2014)

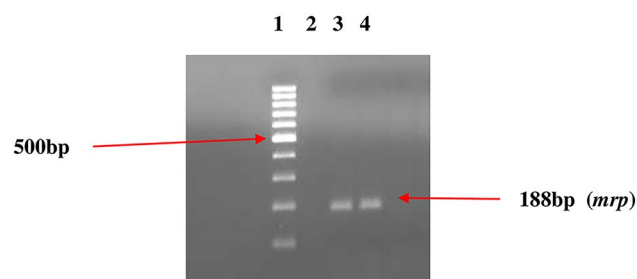


Fig. 3 Detection of *mrp* gene of *S. suis* from pig by PCR. Lane 1: Molecular marker (100 bp DNA ladder), Lane 2: Negative control, Lane 3: Positive control, Lane 4: Test organism

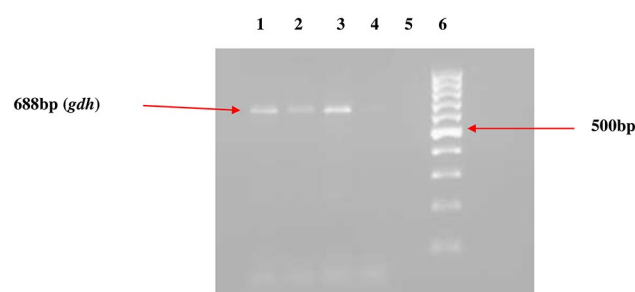


Fig. 4 Detection of *gdh* gene of *S. suis* from pig by PCR. Lane 1–2: Test organism, Lane 3: Positive control, Lane 4–5: Negative control, Lane 6: Molecular marker (100 bp DNA ladder)

reported detection of 15.32% non-typeable *S. suis* isolates through either serotyping or PCR-based *cps* typing. They also opined that these isolates might be acapsular because of a large deletion in the *cps* gene clusters, including the target genes. Gottschalk et al. (2013) observed that most of the non-typable strains of *S. suis* in his study presented high surface hydrophobicity and autoagglutinating characteristics suggesting that these strains are poorly or non-encapsulated. It is also difficult to ascertain if these strains were already non-encapsulated when causing disease, or if they lost their CPS during cultivation. It has also been reported that 34% of isolates belonging to serotype 1/2 or 2 recovered from cases of endocarditis in Japan were non-encapsulated due to deletions and insertions in the genes of the CPS locus (Lakkitjaroen et al. 2011). It was concluded that although the CPS was considered an important virulence factor for *S. suis*, loss of capsule production might be beneficial to *S. suis* in the course of infective

Table 3 Antimicrobial susceptibility of non-typeable *S. suis* isolates (N = 52) from pigs

Antimicrobials used	Number of <i>S. suis</i> isolates		
	Sensitive	Intermediately sensitive	Resistant
Amikacin	43 (82.69)	5 (9.61)	4 (7.69)
Ampicillin	47 (90.38)	3 (5.76)	2 (3.84)
Chloramphenicol	37 (71.15)	5 (9.61)	10 (19.23)
Ciprofloxacin	18 (34.61)	6 (11.53)	28 (53.84)
Clindamycin	5 (9.61)	3 (5.76)	44 (84.61)
Co-trimoxazole	4 (7.69)	5 (9.61)	43 (82.69)
Enrofloxacin	25 (48.07)	9 (17.30)	18 (34.61)
Erythromycin	7 (13.46)	4 (7.69)	41 (78.84)
Gentamicin	29 (55.76)	5 (9.61)	18 (34.61)
Penicillin G	45 (86.53)	4 (7.69)	3 (5.76)
Streptomycin	24 (46.15)	8 (15.38)	20 (38.46)
Tetracycline	3 (5.76)	3 (5.76)	46 (88.46)

Figures in the parentheses indicate the percentage
N number of isolates tested

Table 4 Antimicrobial resistance pattern of 52 non-typeable *S. suis* isolates from pigs

Resistance group	Antimicrobial resistance patterns	Number of isolates
R1	CD-COT	1
R2	CD-TE	3
R3	CIP-HLS	2
R4	C-HLS-TE	1
R5	AK-AMP-COT-TE	1
R6	CD-COT-E-TE	18
R7	CIP-COT-E-EX-HLS-TE	1
R8	C-CD-CIP-COT-E-EX-TE	1
R9	C-CIP-E-EX-HLG-HLS-P	3
R10	CD-CIP-COT-E-EX-HLG-TE	7
R11	CD-CIP-COT-E-HLG-HLS-TE	8
R12	AK-C-CD-CIP-COT-EX-HLS-TE	3
R13	AMP-C-CD-CIP-COT-E-EX-TE	1
R14	C-CD-CIP-COT-E-EX-HLS-TE	2

endocarditis. In fact, non-encapsulated strains were shown to possess not only high adhesion properties to mammalian cells, but also a capacity to form biofilms (Bonifait et al. 2010). Goyette-Desjardins et al. (2014) observed that the sites of isolation of non-typable strains were similar to those of the most important serotypes and emphasized the potential virulence capacity of these non-typable strains in disease causation. *S. suis* infection in humans caused by an unencapsulated strain has also been reported (Kerdsin et al. 2014b).

The present study corroborates with the findings of Princivalli et al. (2009) who also detected similar virulence genes in non-typeable *S. suis* isolates from pigs. To characterize the molecular features of these non-typeable *S. suis* isolates from the diseased and healthy pigs, five virulence-associated genes were investigated. All isolates tested in this study were positive for the *gdh*-PCR assay which specifically detects the glutamate dehydrogenase genes of *S. suis* and the present study corroborated with the findings of Okura et al. (2014). Okwumabua et al. (2001) cloned the gene encoding the glutamate dehydrogenase (GDH) of *S. suis* type 2 and they showed that, like other GDHs, the *S. suis* *gdh* gene appeared to be conserved. All the isolates in the present study were found to be negative for *epf* gene which was consistent with the findings of Zheng et al. (2014) who also found that most of the *S. suis* isolates obtained from China were *epf* negative.

The 52 isolates of *S. suis* tested in this study demonstrated a high genetic diversity, being grouped into five different virulence gene profile. The present study revealed that based on *mrp* gene (*mrp* protein) and *sly* gene (suisysin), the most predominant virulence gene profile was *mrp*⁺ + *sly*⁻ (26.92%,

14 out of 52). The present finding was similar to the findings of Wongsawan et al. (2015) who also observed that this virulence gene profile was more prevalent among the *S. suis* isolates from pigs. As all the isolates in the present study were positive for *gdh* and negative for *epf* genes, the virulence difference among the isolates from clinically healthy pigs and those from diseased pigs may lie in the frequency of the *mrp* and *sly* genes. We have also recorded higher carrier rate of virulence gene profile of *mrp*⁺ + *sly*⁺, *mrp*⁺ + *sly*⁻, and *mrp*⁻ + *sly*⁺ in *S. suis* isolates from diseased pigs compared to healthy pigs, which was consistent with the findings of Silva et al. (2006). Interestingly, all the invasive strains (strains isolated from cases of arthritis and septicemia) in the present study carried *sly* genes whereas only 25% (4 out of 16) isolates from cases of pneumonia carried this *sly* gene which corroborated with the findings of King et al. (2001). It was also observed that 61.53% isolates of *S. suis* carried *arcA* genes in the present study which also corroborated with the findings of Baums et al. (2007) and Matajira et al. (2020) who also observed that *S. suis* isolates from both healthy and diseased pigs were positive for *arcA* genes.

In the absence of effective vaccines to fight against *S. suis*, antimicrobial agents have become increasingly important in treating and controlling the infection of *S. suis* type 2. Of these, β -lactams, tetracyclines, sulphonamides and macrolides are the most common antimicrobials used for the prevention and treatment of streptococcal infection in pig production. But inappropriate use of antimicrobials has led to the development of resistance of *S. suis* to these drugs worldwide (Vela et al. 2005). Prudent use of antimicrobials for *S. suis* is essential to preserve the therapeutic efficacy of broad-spectrum antimicrobials and to minimize selection of resistant *S. suis* strains. It was observed that the most effective antimicrobial against *S. suis* in the present study was ampicillin followed by penicillin. The lowest resistance of *S. suis* isolates for β -lactams was in accordance with the findings of several authors (Wisselink et al. 2006; Zhang et al. 2008), supporting their use as a primary drug for treating *S. suis* infections in pigs. For penicillin, this result (lower resistance) may be due in part to the unique mechanism by which resistance is acquired through modifications in the structure of penicillin-binding proteins (Varela et al. 2013). We have also observed that *S. suis* isolates in the present study showed high frequency of resistance to tetracycline which is consistent with the reports of Varela et al. (2013) who opined that tetracycline resistance among *S. suis* isolates is well documented worldwide. The resistance to tetracyclines in *S. suis* has become a major worldwide problem, closely related to the widespread use of tetracycline in swine production and tetracycline resistance has been considered to be an important cofactor in the selection of resistance to macrolides/lincosamides (Ye et al. 2008). In this study, 46 of the 52 isolates were resistant to tetracycline, 37 of which

were coresistant to macrolides and lincosamides antibiotics which also supported the findings of Ye et al. (2008). In Asia, tetracycline resistance in *S. suis* isolates from both diseased and healthy pigs is common (up to 92%) (Zhang et al. 2008). High levels of tetracycline resistance (up to 90%) have also been reported among *S. suis* isolates from diseased and clinically healthy persons (Hoa et al. 2011). Resistance to macrolide antibiotic has been observed among *S. suis* isolates from different pig producing countries of the world. In Quebec, Canada, up to 82% of *S. suis* isolates were resistant to erythromycin (Cantin et al. 1992). In the USA, a retrospective study of 256 cases over a 4-year period (1985–1989) reported that 45% of 65 *S. suis* isolates were resistant to erythromycin (Reams et al. 1993). In Spain, more than 87% of *S. suis* isolates were resistant to tetracycline and oxytetracycline, sulphadimethoxine and sulphathiazole (Vela et al. 2005). Macrolide resistance is also prevalent in *S. suis* isolates from clinically healthy pigs, where rates up to 67% have been reported (Zhang et al. 2008). Many antimicrobials used in veterinary practice belong to the same classes as antimicrobials used to treat humans (McEwen and Fedorka-Cray 2002). The use of antimicrobials in livestock may allow transfer of resistant bacteria among animals and resistance genes may be transferred to both commensal organisms and animal or human pathogens (McEwen and Fedorka-Cray 2002). It has been suggested that *S. suis* may act as an antibiotic resistance reservoir contributing to the spread of resistance genes to streptococcal human pathogens (Palmieri et al. 2011). Thus, occurrence of relatively high levels of resistance of *S. suis* to some antimicrobials (e.g. lincosamides, macrolides, tetracyclines, and sulphonamides) as observed in the present study may represent a human health concern.

This is perhaps the first study reporting virulence gene profile and antimicrobial resistance of non-typeable *S. suis* isolated from Indian piggeries. The results obtained in this study are a cause of concern as the virulence gene profiles of non-typeable *S. suis* circulating in Indian pigs are also reported in *S. suis* serotype 2 from both pigs and humans from different pig producing countries of the world. Increasing resistance of *S. suis* isolates to commonly used antimicrobials as observed in the present study is also another area of concern. Increased awareness of *S. suis* in both human and veterinary diagnostic laboratories and further establishment of virulence gene profile of circulating strains in a particular region/area will contribute to our knowledge of this pathogen, especially in regions where complete and/or recent data is lacking.

In conclusion, non-typeable *S. suis* isolates from Indian pigs were found to carry virulence-associated genes such as *mrp*, *sly*, *arcA*, and *gdh* whereas all isolates were found negative for *epf* gene. The study revealed that the most predominant virulence gene profile found in healthy pigs was *mrp*⁺ + *sly*⁺ + *arcA*⁺ + *gdh*⁺ + *epf*⁺ whereas the most predominant

virulence gene profile recorded in diseased pigs was *mrp*⁺ + *sly*⁺ + *arcA*⁺ + *gdh*⁺ + *epf*⁺. High frequency of resistance was observed for tetracycline followed by clindamycin, cotrimoxazole, and erythromycin with 88.46% isolates being multi-drug resistant.

Declarations

Conflict of interest The authors declare no competing interests.

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